

TechNova Solutions Pvt. Ltd.

Enterprise Knowledge Base

Product Documentation

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Version	Date	Author	Changes
1.0	July 2026	Product Team	Initial Release

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1. Purpose

This document provides an overview of TechNova's enterprise software products, their architecture, supported features, user roles, APIs, deployment standards, and maintenance procedures. It serves as the primary reference for engineers, product managers, and support teams.

2. Scope

This document applies to:

- Product Managers
 - Software Engineers
 - QA Engineers
 - DevOps Engineers
 - AI Engineers
 - Customer Support Teams
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3. Product Portfolio

TechNova develops the following enterprise products:

HRConnect

Human Resource Management System

Features:

- Employee Management
- Leave Management
- Payroll
- Attendance
- Performance Reviews

SalesVision

Customer Relationship Management (CRM)

Features:

- Lead Tracking
 - Opportunity Management
 - Sales Dashboard
 - Customer Analytics
 - Revenue Forecasting
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InsightAI

Enterprise AI Assistant

Features:

- Document Search
 - Knowledge Retrieval
 - RAG-based Question Answering
 - Report Generation
 - Internal Chat Assistant
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OpsMonitor

Infrastructure Monitoring Platform

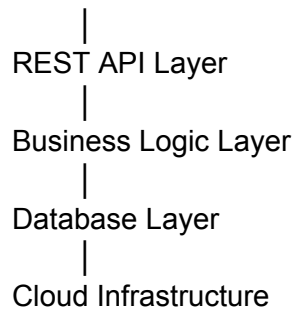
Features:

- Server Monitoring
 - Application Monitoring
 - Alert Management
 - Performance Dashboard
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4. Product Architecture

Each product follows a layered architecture:

Client Applications



This architecture ensures scalability, maintainability, and security.

5. User Roles

Role	Responsibilities
Administrator	Full system access
Manager	Team management and reporting
Employee	Daily operational activities
Support Engineer	Incident resolution
Auditor	Read-only access for compliance

Access is granted based on the principle of least privilege.

6. Core Modules

Each product includes:

- Authentication
- Dashboard
- Notifications
- Reporting
- Audit Logs
- User Management
- Role-Based Access Control (RBAC)
- Search Functionality

Modules are designed as reusable components across products.

7. Authentication & Authorization

Authentication methods:

- Company Email Login
- Multi-Factor Authentication (MFA)
- Single Sign-On (SSO)

Authorization is enforced through Role-Based Access Control (RBAC), ensuring users can access only the resources required for their roles.

8. API Standards

All APIs follow REST principles.

Standards include:

- JSON request and response format
- HTTPS for all communications
- Versioned endpoints (e.g., `/api/v1/`)
- Standard HTTP status codes
- Token-based authentication (JWT)

API documentation is maintained using OpenAPI specifications.

9. Database Design

Supported databases:

- PostgreSQL
- MySQL
- MongoDB (selected services)

Guidelines:

- Normalize transactional data.
- Index frequently queried columns.

- Encrypt sensitive information.
 - Schedule regular backups.
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10. Performance Guidelines

Applications should meet the following targets:

Metric	Target
Average API Response Time	< 500 ms
Page Load Time	< 2 seconds
Availability	99.9%
Database Query Time	< 200 ms

Performance is continuously monitored through automated dashboards.

11. Logging & Monitoring

All products generate logs for:

- User Login Activity
- API Requests
- Application Errors
- Database Events
- Security Alerts

Monitoring tools include:

- Prometheus
- Grafana
- ELK Stack

Logs are retained for 90 days unless regulatory requirements specify otherwise.

12. Release Management

Software releases follow this process:

- 1. Feature Development
- 2. Code Review
- 3. Automated Testing
- 4. User Acceptance Testing (UAT)
- 5. Release Approval
- 6. Production Deployment
- 7. Post-Release Monitoring

Emergency hotfixes follow an expedited approval process.

13. Backup & Recovery

The company performs:

- Daily Incremental Backups
- Weekly Full Backups
- Monthly Recovery Testing

Recovery objectives:

Objective	Target
Recovery Time Objective (RTO)	4 Hours
Recovery Point Objective (RPO)	1 Hour

14. Troubleshooting

Common issues and recommended actions:

Issue	Resolution
User cannot log in	Verify credentials, MFA status, and account lock
Slow application	Check server load and database performance
API failure	Review API logs and restart affected services

Missing reports

Confirm data synchronization and permissions

Email notifications not sent

Check SMTP service and mail queue

Escalate unresolved issues to the Product Engineering Team.

15. Product Roadmap

Planned enhancements:

- AI-powered analytics
- Mobile application improvements
- Advanced workflow automation
- Voice-enabled search
- Predictive reporting
- Enhanced cybersecurity features

Roadmaps are reviewed every quarter.

16. Frequently Asked Questions

Which authentication method is mandatory?

All users must authenticate using company credentials with Multi-Factor Authentication (MFA).

What is the target system availability?

The company aims for **99.9% uptime** across production environments.

Which database is preferred?

PostgreSQL is the standard relational database for most enterprise applications.

How often are backups performed?

Incremental backups are performed daily, with full backups every week.

How should API changes be documented?

All API updates must be reflected in the OpenAPI documentation before release.

Product Standards Summary

Area	Standard
API Protocol	REST
Authentication	SSO + MFA
Database	PostgreSQL
Logging	ELK Stack
Monitoring	Prometheus & Grafana
Availability	99.9%
Backup	Daily Incremental
Deployment	CI/CD Pipeline

Related Documents

- Engineering Handbook
- AI Department Handbook
- Information Security Handbook
- IT Helpdesk Guide